INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April - May, 2026

Degree: **B.C.A.** Semester: **Second**

Stream: **B.C.A.**

Paper Code: **BCACC203**

Paper Name: **Computer Architecture**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. What is the binary representation of 0.125? 2
2. What is the main task in first pass of assembler? 2
3. Explain instruction formats. 2
4. What is SISD? 2
5. In which method of number representation there are two possible representations for the value "0"? Explain your answer. 2
6. Define Direct Memory Access. 2
7. What is speedup in pipeline? 2
8. Determine the total number of clock cycles required to process 200 tasks in a six-segment pipeline. 2
9. Convert the decimal number 179 into its equivalent base-7 representation. 2
10. Define Immediate mode. 2
11. Explain, with proper reasoning, which set of logic gates are used in the implementation of a binary multiplier circuit. 2
12. Define SIMD. 2

13. In which addressing mode is the operand value specified in the instruction? Explain your answer. 2

14. What is seek time? 2

15. Define bus request. 2

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Illustrate register transfer control function with the help of a block diagram. 5

17. What is the difference between a direct and an indirect address instruction? How many references to memory are needed for each type of instruction to bring an operand into a processor register? 5

18. Evaluate X = (A + B) * (C + D) using one-address instruction. 5

19. Summarize the concept of 4 segments pipelining with space-time diagram. 5

20. Starting from an initial value of R =11011101, determine the sequence of binary values in R after a logical shift-left, followed by a circular shift-right, followed by a logical shift-right and a circular shift-left. 5

21. Write the difference between hardware control program and micro programmed control program. 5

22. What is the difference between a software interrupt and subroutine call? 5

23. What are the three different registers used in DMA Controller and what are their activities? 5

24. Register A holds the 8-bit binary 11011001. Determine the B operand and the logic microoperation to be performed in order to change the value in A to: 5

    a) 01101101
    b) 11111101

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Describe the Register Transfer Language (RTL). Explain various RTL symbols and write RTL statements for the following operations: **10**
    a) Transfer contents of R5 to R1 only if control signal P = 1.
    b) Increment register R2 by 1.
    c) Subtract R3 from R4 and store the result in R4.
    d) Perform OR operation between R1 and R2 and store the result in R1.
    e) Perform circular shift right on register R6.

26. Evaluate the step by step multiplication process using Booth's algorithm for ((+15) times (-13)). **10**

27. Illustrate memory hierarchy in a computer system. **10**

28. Describe the concept of mapping in cache memory? Briefly explain all the 3 types of mapping procedure. **10**

29. Explain the flowchart of divide operation. **10**

30. Summarize the flowchart for the interrupt cycle and explain it. **10**

31. Present a diagram for control unit of basic computer and describe it in details. **10**

32. A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers. **10**
    a) How many selection inputs are there in each multiplexer?
    b) What size of multiplexers are needed?
    c) How many multiplexers are there in the bus?

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